

Natural Language Processing

Why NLP?

- Huge amount of data
 - Internet at least 2.5 billion pages
- Application for processing large amount of texts
 - Requires NLP expertise
- Application
 - Classify text into category
 - Index and search large text
- Automatic translation
 - Speech understanding
 - Understand phone conversation
- Information extraction
 - Extract useful information from resumes
- Automatic summarisation
 - Question answering
- Knowledge acquisition
 - Text generation /dialogs

Natural?

- Natural language
 - Refers to the language spoken by people e.g. English, Japanese..... As opposed to artificial languages like C++, Java etc.
- Natural language processing
 - Applications that deal with natural language in a way or another
- Computational linguistics
 - Doing linguistics on computers
 - More on the linguistic side than NLP but closely related

Why NLP?

- Kjtmm mmmnn
- Mmjhsdjhm
- mmmmb

Computers lack knowledge

- Computers “see” text in English in the same way you have seen in the previous text
- People have no trouble understanding language
 - Commonsense knowledge
 - Reasoning capacity
 - Experience
- Computers have
 - No commonsense knowledge
 - No reasoning capacity unless we teach

Linguistics levels of analysis

- Speech
- Written language
 - Phonology
 - Morphology
 - Syntax
 - semantics

Issues in syntax

- “the dog ate my homework” who did what?
- Identify the part of speech
 - Dog = noun; ate = verb; homework = noun;
 - English POS tagging; 95% can be improved
- Identify collocations
 - Mother in law, hot dog
- The sentence is syntactically correct...
- What is POS tagging?

Issues in syntax

- Shallow parsing:
 - “the dog chased the bear”
 - “the dog”, “chased” “the bear”
 - Subject /predicate
 - Identify basic structure
 - NP [the dog] VP [chased the bear]
 - Shallow parsing on new language
 - Shallow parsing with little training data

Full parsing

Full parsing

More issues in Syntax

- Anaphora Resolution:
- “The dog entered my room. It scared me”
- Proposition attachment
- “I saw the man in the park with a telescope”

Issues in semantics

- Understand language; how?
- “plant”
- Important of semantics?
 - Machine translation: wrong translation
 - Information retrieval: wrong information
 - Anaphora resolution: wrong reference

Issues in semantics

- How to learn the meaning of words
 - From dictionaries
 - Context is becoming important!! How???
-
- The plant was close to the farm of animals
 - The sea flora consists in 1000 different plant species

Issues in semantics

- Learn from annotated example:
 - Assume 100 examples containing “plant” previously tagged by a human
 - Train a learning algorithm
 - Precision in the range 60
 - How to choose the learning algorithm?
 - How to obtain the 100 tagged examples
 - Hidden markov model/NN

Issues in Learning semantics

- Learning
 - Assume a large amount of annotated data = training
 - Assume a new text annotated = test
- Learn from previous experience = training to classify new data = test
- Decision trees, memory based learning, neural networks
 - Machine learning

Issues in information extraction

- “there was a group of about 8 - 9 people close to the entrance on highway 64”
- Who?
- Where”
- Extract information
- Detect new patterns

Issues in information retrieval

- General model:
 - A huge collection of text
 - A query
- Task: find documents that are relevant to the given query
- How? Create an index
- Examples: google

Issues in information retrieve

- Index meaning
- Search for plant

example

Issue

Issues in MT

- Text to text
- Speech to speech

The model